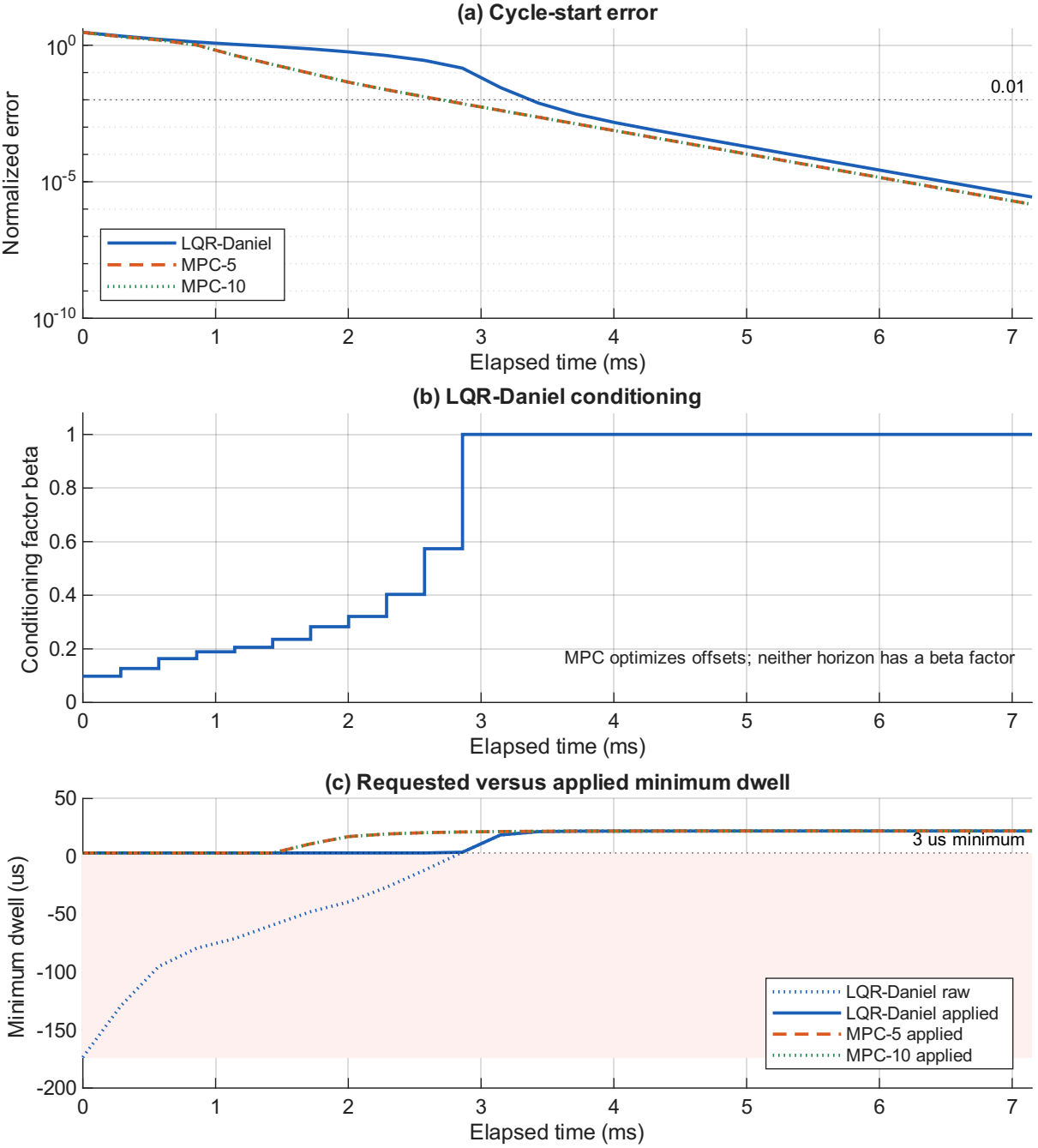
(a) Cycle-start error
Normalized error
10^{0}
10^{-5}
10^{-10}
0.01
LQR-Daniel
MPC-5
MPC-10
0
1
2
3
4
5
6
7
Elapsed time (ms)
(b) LQR-Daniel conditioning
Conditioning factor beta
1
0.8
0.6
0.4
0.2
0
MPC optimizes offsets; neither horizon has a beta factor
Elapsed time (ms)
(c) Requested versus applied minimum dwell
Minimum dwell (us)
50
0
-50
-100
-150
-200
3 us minimum
LQR-Daniel raw
LQR-Daniel applied
MPC-5 applied
MPC-10 applied
Elapsed time (ms)